

Semiconductor Capital Equipment

3Q'26 WFE update: 2026/27 Bull Case is the Base Case

We revise up our 2026/27/28 WFE market forecast to +39%/+37%/+14% as the industry blows post constraints. We think the debate starts to shift from pace of growth (WFE) to end-demand (compute/GW).

What's Changed? We revise up our 2026 forecast from \$155bn (up 32% y/y) to \$163bn (up 39%), our 2027 forecast comes up from \$201bn (up 31%) to \$223bn (up 37%) and our 2028 forecast comes up from \$227bn (up 13%) to \$254bn (up 14%). Our largest revision is in DRAM and Leading Logic, where we continue to revise up numbers higher and higher for DRAM while reflecting a broadening of spending in Leading Logic.

2026/27 Bull Case is the Base Case. Since [Dec 1st](#), we have raised our 2026 WFE forecast by 26% and our 2027 forecast by 54%. We have seen unprecedented DRAM pull-ins into 2026, driving a \$17bn (47%) increase in our DRAM WFE forecast. Our 2027 outlook has also strengthened as key elements of our bull case have begun to materialize, including Intel spending of \$20bn+, approx. 500kwpm of DRAM greenfield capacity, and a restart in NAND greenfield investment. On a quarterly basis, we now model the industry reaching a \$200bn annualized run rate by DecQ 2026.

Shifting the SPE Debate from pace of growth to end-demand (compute/GW). Our view that the "[WFE cycle has longevity](#)" was premised on demand (cleanroom) and supply (supply chain) constraints limiting near-term upside while extending the medium-term opportunity. Quite frankly, the industry now looks set to blow past both constraints. Based on our revised 2026/27 forecasts, we see DRAM bit supply growth reaching 38% in 2027 (vs. 24% in 2025) and 2nm capacity reaching 335kwpm/500kwpm in 2027/28 (vs. 77kwpm in 2025). We have underestimated both the strength of demand and the pace of supply additions - this is now our 5th WFE forecast revision YTD, and we continue to see risks skewed to the upside. However, our \$254bn WFE forecast for 2028 implies a meaningful growth deceleration off a much higher 2027 base, from 37% y/y growth in 2027 to 14% in 2028. WFE would need to exceed \$300bn in 2028 to sustain the current growth rate. Importantly, this is despite our 2027 DRAM WFE forecast remaining below what SPE companies appear to imply (\$75bn+ vs. MSe at \$70bn). Industry growth of \$100bn+ from 2025 to 2027 more closely aligns with the compute (GW) deployment we're seeing (approx. 30GW of WFE vs 2027/28 GW of 29/36), based on our analysis of [WFE per GW](#) and our [Internet Team's Forecast of industry GW deployment](#). We're not bearish; we're pleasantly surprised. The industry has gone from a ~\$125bn annualized run rate to ~\$200bn in just four quarters (4Q25 to 4Q26). For our stock views, see [Thoughts on stocks](#).

MORGAN STANLEY & CO. LLC		
Shane Brett		
Equity Analyst		
Shane.Brett@morganstanley.com		+1 212 761-1022
MORGAN STANLEY & CO. INTERNATIONAL PLC+		
Lee Simpson		
Equity Analyst		
Lee.Simpson@morganstanley.com		+44 20 7425-3378
Nigel van Putten		
Equity Analyst		
Nigel.Putten@morganstanley.com		+44 20 7425-2803
MORGAN STANLEY TAIWAN LIMITED+		
Charlie Chan		
Equity Analyst		
Charlie.Chan@morganstanley.com		+886 2 2730-1725
MORGAN STANLEY MUFG SECURITIES CO., LTD.+		
Suzune Tamura, CFA		
Equity Analyst		
Suzune.Tamura@morganstanleymufg.com		+81 3 6836-8891
MORGAN STANLEY & CO. LLC		
Joseph Moore		
Equity Analyst		
Joseph.Moore@morganstanley.com		+1 212 761-7516
MORGAN STANLEY ASIA LIMITED+		
Henry Zhao		
Research Associate		
Henry.Zhao@morganstanley.com		+852 2239-7731
MORGAN STANLEY & CO. INTERNATIONAL PLC+		
Amelia M Scicluna		
Research Associate		
Amelia.Scicluna@morganstanley.com		+44 20 7425-6694
MORGAN STANLEY & CO. LLC		
Mason Wayne		
Research Associate		
Mason.Wayne@morganstanley.com		+1 212 761-6012
Ella Tulchinsky		
Research Associate		
Ella.Tulchinsky@morganstanley.com		+1 212 761-2222
Nicole Kozhukhov		
Research Associate		
Nicole.Kozhukhov@morganstanley.com		+1 212 761-1636

SEMICONDUCTOR CAPITAL EQUIPMENT	
North America	
Industry View	In-Line

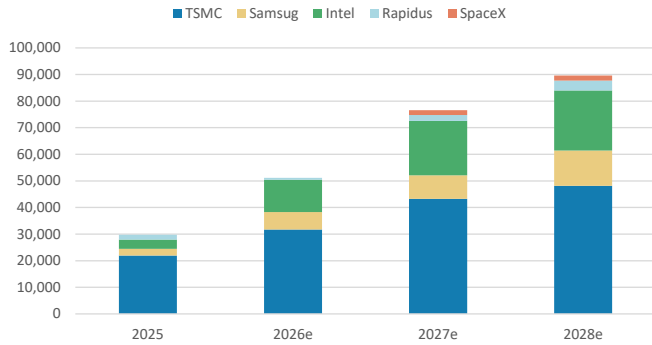
Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

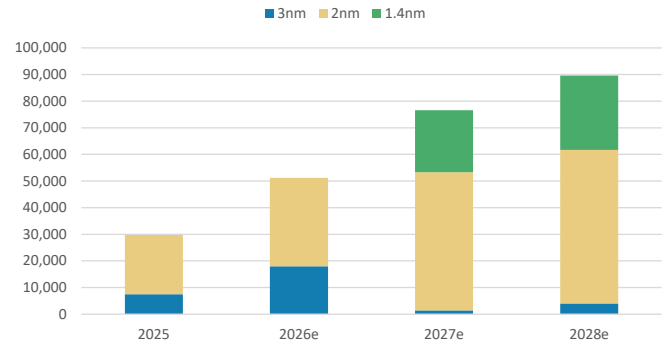
Our story in 6 charts

Exhibit 1: We model Intel to be a meaningful No.2 spender in 2027 leading logic WFE...



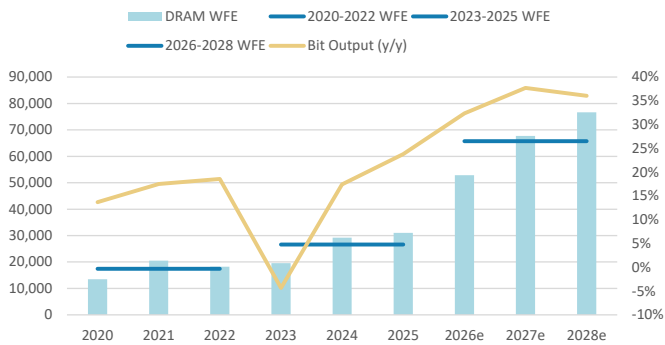
Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 2: ...and forecast industry spend of approx. \$50bn towards 2nm



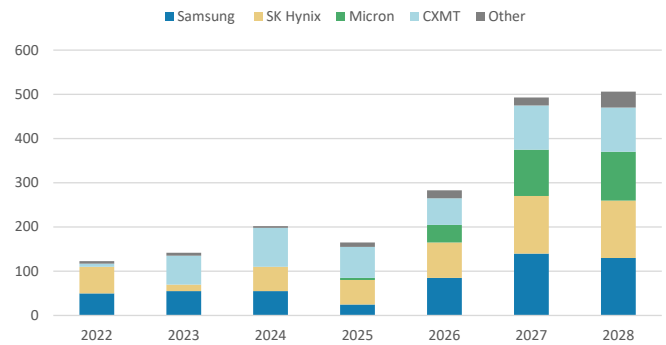
Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 3: SPE Cos are implying \$75bn+ of DRAM WFE, but we're landing at \$70bn, which still generates 38% bit supply growth.



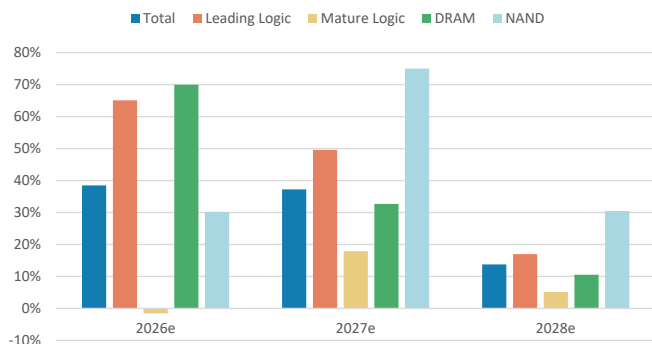
Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 4: We see approx. 500kwpm of DAM greenfield in 2027/28



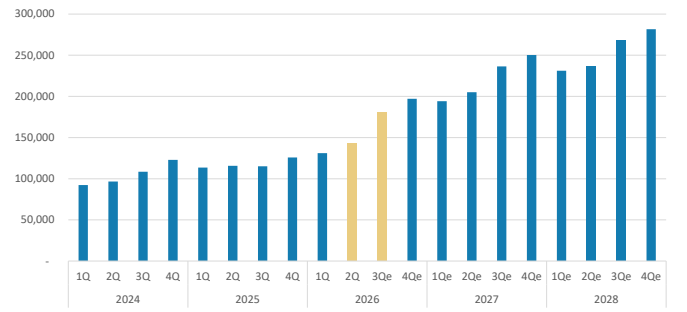
Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 5: We continue to model NAND to be the fastest grower in 2027, off a slightly lower base.



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 6: We're pleasantly surprised that the industry has gone from a \$125bn run rate to a \$200bn run rate in 4 quarters



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

WFE Forecast Table

Exhibit 7:

(\$mn)	2022	2023	2024	2025	2026e	2027e	2028e
WFE Revenue	96,222	96,008	105,057	117,530	162,782	223,360	254,124
Semi Revenue	573,978	526,820	630,549	791,061	1,718,383	2,222,732	2,222,732
Semi Capex	187,982	176,924	198,540	215,972	330,906	400,806	400,597
Y/Y Change %							
WFE Revenue	6%	0%	9%	12%	39%	37%	14%
Semi Revenue	4%	-8%	20%	25%	117%	29%	0%
Semi Capex	19%	-6%	12%	9%	53%	21%	0%
Metrics							
WFE Intensity	17%	18%	17%	15%	9%	10%	11%
WFE% of Total Capex	51%	54%	53%	54%	49%	56%	63%
Semi Capital Intensity	33%	34%	31%	27%	19%	18%	18%
3rd Party Sources							
Semi	94,100	95,610	104,270	116,900	143,900	175,300	200,000
Gartner	101,101	102,820	111,777	124,452	148,052	166,085	164,697
WFE by segment							
Foundry/Logic	59,180	70,023	69,192	75,307	94,818	127,995	143,672
Leading Logic		27,500	24,500	31,000	51,189	76,572	89,589
Mature Logic		42,523	44,692	44,307	43,629	51,422	54,083
Memory	35,854	24,794	35,228	41,600	66,515	94,048	108,722
DRAM	18,256	19,579	29,244	31,083	52,827	70,086	77,456
NAND	17,598	5,215	5,985	10,517	13,689	23,963	31,266
Other	1,188	1,192	637	622	1,449	1,316	1,730
Y/Y Change %							
Foundry/Logic	18%	18%	-1%	9%	26%	35%	12%
Leading Logic			-11%	27%	65%	50%	17%
Mature Logic			5%	-1%	-2%	18%	5%
Memory	-10%	-31%	42%	18%	60%	41%	16%
DRAM	-12%	7%	49%	6%	70%	33%	11%
NAND	-8%	-70%	15%	76%	30%	75%	30%
Other	19%	0%	-47%	-2%	133%	-9%	31%
% of Total							
Foundry/Logic	62%	73%	66%	64%	58%	57%	57%
Leading Logic		29%	23%	26%	31%	34%	35%
Mature Logic		44%	43%	38%	27%	23%	21%
Memory	37%	26%	34%	35%	41%	42%	43%
DRAM	19%	20%	28%	26%	32%	31%	30%
NAND	18%	5%	6%	9%	8%	11%	12%
Other	1%	1%	1%	1%	1%	1%	1%
WFE Intensity							
Semi ex-Memory revenue	444,153	434,511	465,032	568,550	737,371	856,197	856,197
Memory revenue	129,825	92,309	165,516	222,512	981,012	1,366,535	1,366,535
DRAM	77,769	51,945	94,860	150,598	657,914	844,324	844,324
NAND	47,109	36,104	66,427	67,685	318,868	517,982	517,982
WFE Intensity							
Foundry Logic	13%	16%	15%	13%	13%	15%	17%
Memory	28%	27%	21%	19%	7%	7%	8%
DRAM	23%	38%	31%	21%	8%	8%	9%
NAND	37%	14%	9%	16%	4%	5%	6%
% of Memory WFE							
DRAM	51%	79%	83%	75%	79%	75%	71%
NAND	49%	21%	17%	25%	21%	25%	29%

Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 8:

(\$mn)	2022	2023	2024	2025	2026e	2027e	2028e
WFE by Region							
Total	96,222	96,008	105,057	117,530	162,782	223,360	254,124
North America	10,239	11,347	13,657	10,578	15,464	22,336	25,412
Europe	5,982	5,742	4,202	2,351	2,767	3,350	3,812
Japan	7,244	6,439	7,354	8,815	11,395	16,082	18,551
Korea	19,656	18,966	18,910	23,506	39,068	53,606	59,211
Taiwan	26,566	19,501	15,233	27,620	43,951	62,541	71,155
China	22,104	31,681	42,548	41,723	45,579	55,840	68,613
Other	4,431	2,334	3,152	2,938	4,558	9,604	7,370
WFE by Region (Y/Y)							
North America	32%	11%	20%	-23%	46%	44%	14%
Europe	108%	-4%	-27%	-44%	18%	21%	14%
Japan	5%	-11%	14%	20%	29%	41%	15%
Korea	-14%	-4%	0%	24%	66%	37%	10%
Taiwan	13%	-27%	-22%	81%	59%	42%	14%
China	-10%	43%	34%	-2%	9%	23%	23%
Other	73%	-47%	35%	-7%	55%	111%	-23%
WFE by Region (% of Total)							
North America	11%	12%	13%	9%	10%	10%	10%
Europe	6%	6%	4%	2%	2%	2%	2%
Japan	8%	7%	7%	8%	7%	7%	7%
Korea	20%	20%	18%	20%	24%	24%	23%
Taiwan	28%	20%	15%	24%	27%	28%	28%
China	23%	33%	41%	36%	28%	25%	27%
Other	5%	2%	3%	3%	3%	4%	3%

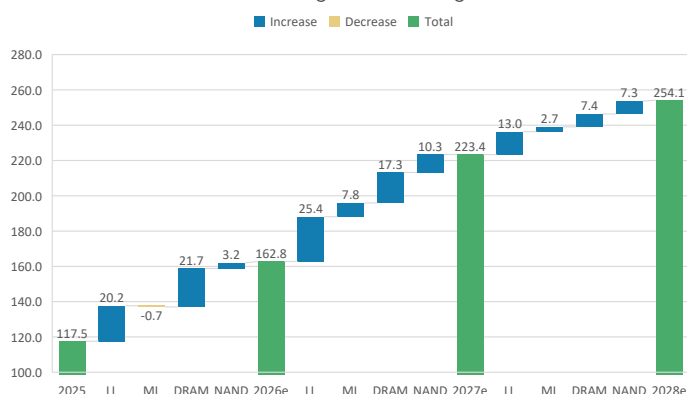
Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

How do our numbers change?

What's Changed? We revise up our 2026 forecast from \$155bn (up 32% y/y) to \$163bn (up 39%), our 2027 forecast comes up from \$201bn (up 31%) to \$223bn (up 37%) and our 2028 forecast comes up from \$227bn (up 13%) to \$254bn (up 14%)

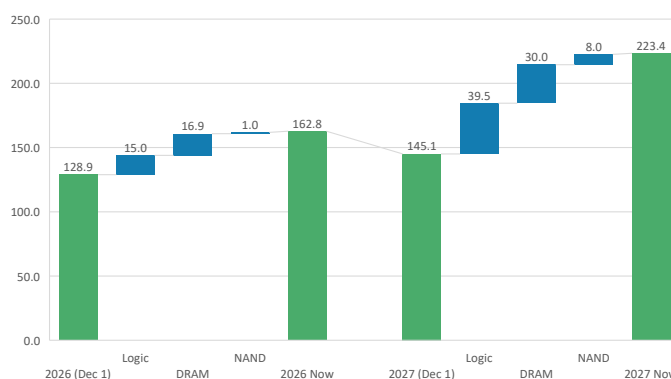
Our View: Our largest revision is in DRAM and Leading Logic, where we continue to revise up numbers higher and higher for DRAM while reflecting a broadening of spending in Leading Logic.

Exhibit 9: We now model high-30% WFE growth in 2026/27



Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 10: Our 2026/27 WFE forecasts have come up 26%/54% since Dec 1st.



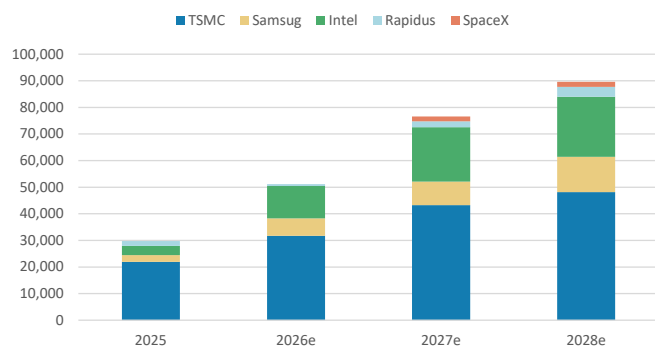
Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Leading Logic: Two very strong years of growth

We revise up our 2026/27 Logic WFE forecast from \$89bn/\$114bn to \$95bn/\$128bn. We introduce our leading edge logic wafer model and forecast the following:

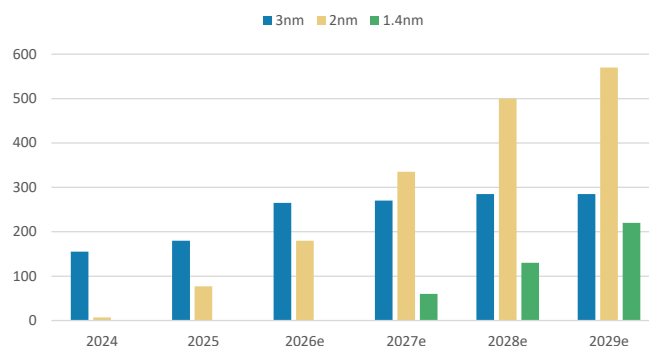
- WFE: We forecast leading logic WFE growth of 72% in 2026 and 50% in 2027. We forecast mature logic WFE to decline 3% in 2026 and grow 15% in 2027. We model mix to reverse from 39%/61% in 2025 to 63%/37% in 2028 (leading vs mature).
- 3nm: We model wafer capacity to expand from 180kwpm in 2025 to 265kwpm in 2026 but see marginal increases during 2027-29.
- 2nm: We model wafer capacity of 180kwpm in 2027 and 335kwpm in 2028, with adds across TSMC, Intel and Samsung.
- 1.4nm: We model wafer capacity of 60kwpm in 2027 and 130kwpm in 2028 led by TSMC and Intel.

Exhibit 11: TSMC should be the biggest contributor to growth in 2027, closely followed by Intel



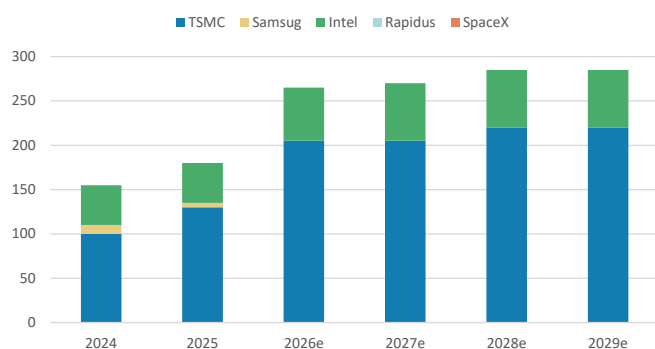
Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 12: 2nm capacity should surpass peak 3nm in 2027 led by adds across all 5 players



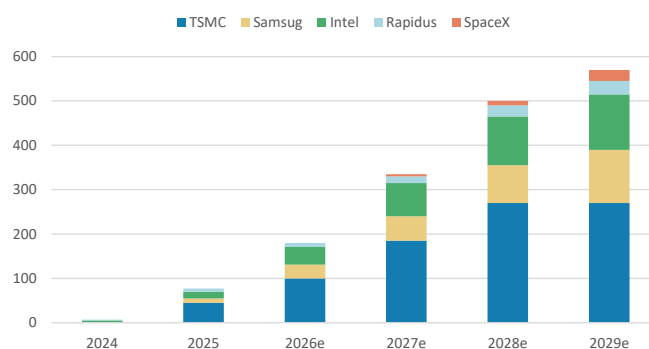
Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 13: After meaningful adds in 2026, we expect 3nm to flatline from 2027



Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 14: We expect 2nm capacity to surpass peak 3nm in 2027 and continue to increase



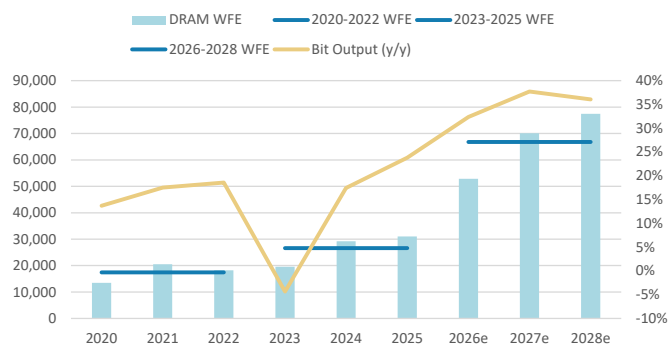
Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

DRAM: \$120bn+ over 2026/27

We revise up our 2026/27 DRAM WFE forecast from \$49bn/\$62bn to \$53bn/\$70bn. Our changes are as follows:

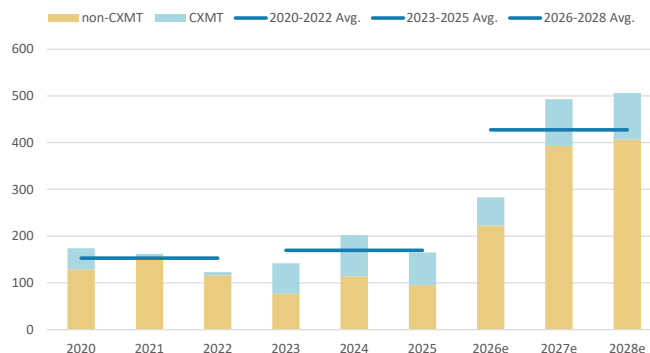
- **Greenfield:** We adjust our 2026/2027 forecast from 268/395kwpm to 283/493kwpm. Our 2026 revision is reflective of small tweaks at Hynix and Micron. Our 2027 revision is across the board but with the biggest revisions at Samsung and Hynix.
- **Bit Output:** Our revised wafer forecasts lead to our 2026/2027 bit supply growth forecasts being revised up from 29%/33% to 32%/38%.

Exhibit 15: We model annual DRAM WFE to average \$67bn during 2026-28, up from \$27bn in 2023-25



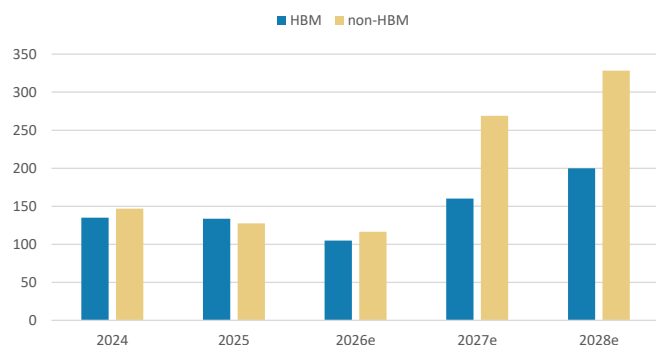
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 16: We model annual DRAM greenfield adds to average 427kwpm during 2026-28, up from 170kwpm in 2023-25



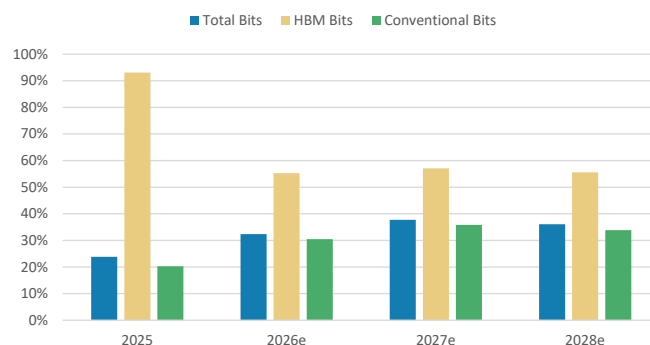
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 17: We model non-HBM wafer adds to track ahead of HBM wafer adds from 2027



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 18: We model 2026/2027 HBM bit supply growth of 55%/57% vs conventional bit supply growth of 30%/36%.



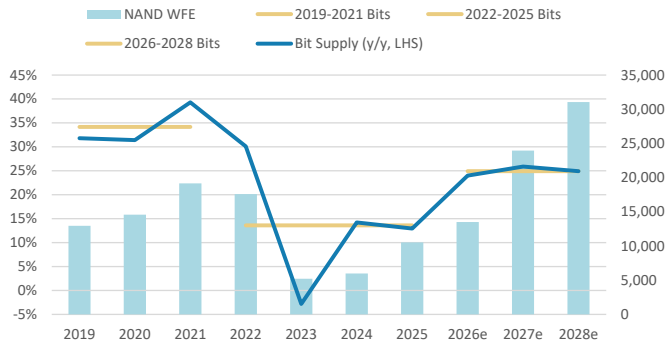
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

NAND: Minimal changes

We fine-tune our 2026/27 NAND WFE forecast from \$15.6bn/\$24.0bn to \$13.7bn/\$24.0bn. Our changes are as follows;

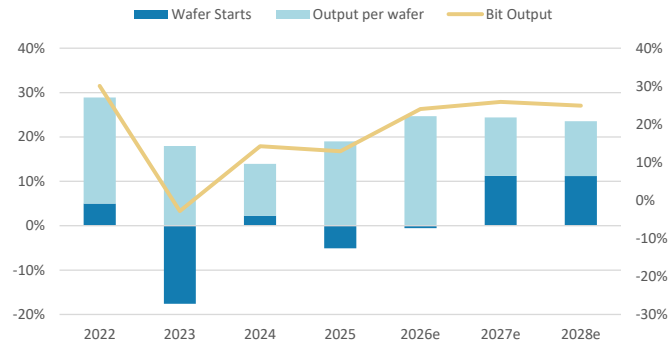
- WFE: our greenfield forecasts for 2026/27 are largely unchanged but we slightly lower our brownfield forecast for 2026. YMTC continues to account for approx. half of our greenfield adds in 2026/27.
- Bit Output: Given our lower 2026 NAND WFE forecast, our bit supply growth forecast comes down from 24%/31% to 24%/26%.

Exhibit 19: We model NAND bit supply to flat-line in the mid-20s due to lower 2026 NAND WFE.



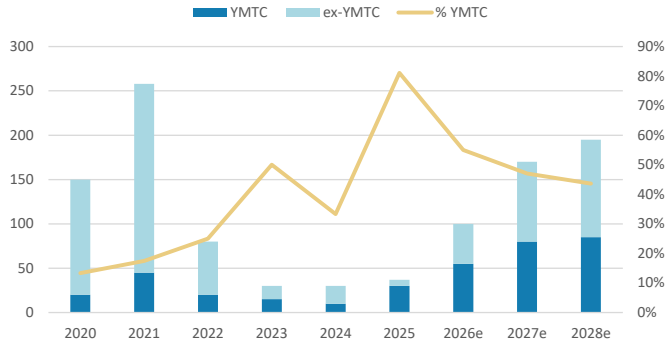
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 20: Wafer adds should contribute to bit supply growth from 2027



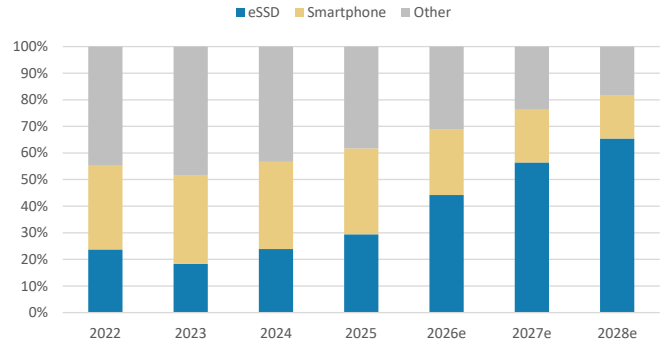
Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 21: We continue to model YMTC to make up approx. 50% of capacity adds



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 22: We model mix of eSSD to increase from 29% in 2025 to 65% in 2028



Source: TrendForce, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Thought #1: Leading Logic

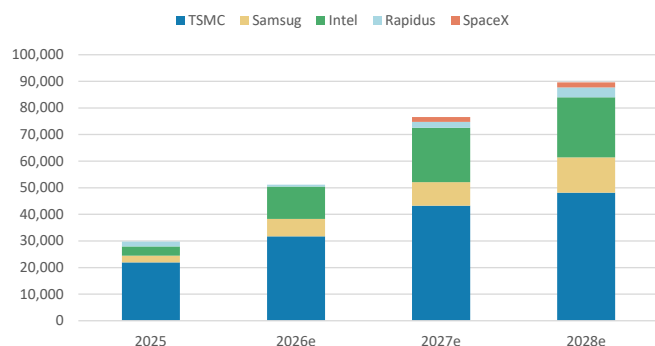
Our View: We see leading logic WFE growth remaining strong into 2027 driven by TSMC accelerating 2nm adds and Intel becoming a meaningful No. 2 spender.

The Numbers: We model leading logic WFE growth of 65%/50% in 2026/27 and mature logic growth of -2%/+18%.

We see the following dynamics through 2028;

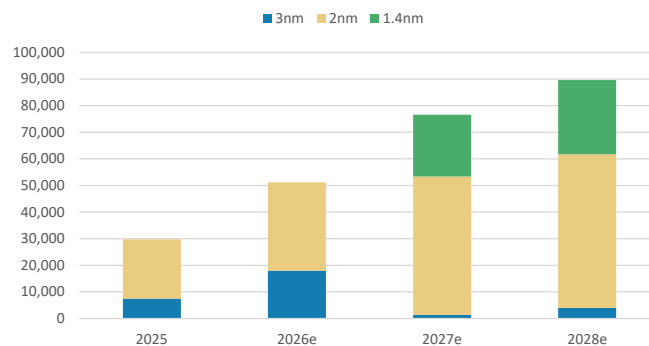
- 2026 \$51bn (+65% y/y): We see similar growth between 3nm and 2nm, with spending driven by TSMC, Samsung, and Intel.
- 2027 \$76bn (+50% y/y): We see 2nm as the biggest growth driver with initial 1.4nm wafer additions. We see TSMC and Intel as the biggest growth drivers as we see TSMC spend \$43.3bn and Intel spend \$20.5bn.
- 2028 \$90bn (+17% y/y): We conservatively model TSMC's and Intel's spending growth to decelerate, slightly offset by a pick-up at Samsung. We don't reflect a meaningful contribution from SpaceX (\$2bn).

Exhibit 23: We model Intel to be a meaningful No.2 spender in 2027.



Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

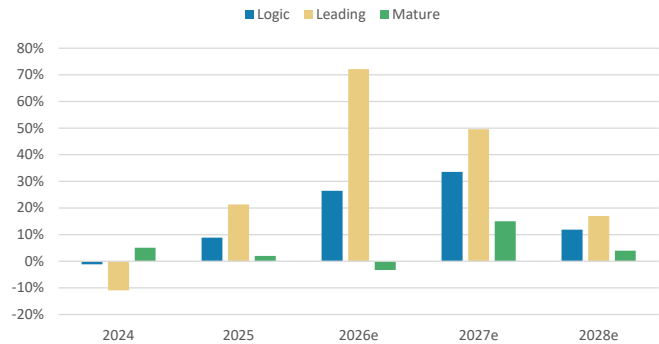
Exhibit 24: We model approx. \$50bn of 2nm spend in 2027



Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

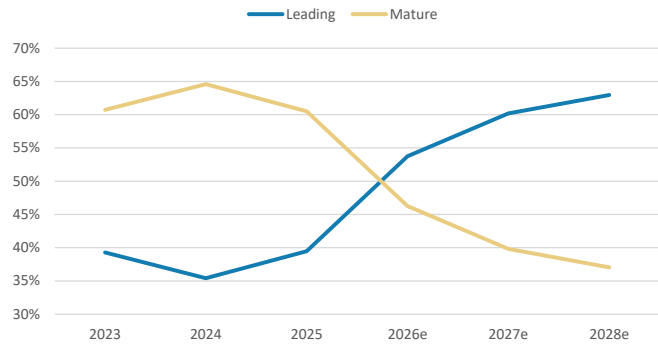
Waiting for mature node acceleration. Contrary to the leading edge where we see a broadening of spenders and 2nm being a huge node, our forecast on mature node is substantially more conservative as we model spending to have been roughly flat from 2023-26 with a slight pick up (+18%) in 2027.

Exhibit 25: We model leading logic WFE growth to outpace total logic WFE through 2028



Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 26: We model leading vs mature split to reverse from 35%/65% in 2024 to 63%/37% in 2028



Source: Morgan Stanley Research. e = Morgan Stanley Research estimates

Thought #2: DRAM

Our View: SPE companies have called for DRAM WFE to track ahead of total WFE growth in 2027 which would imply \$75bn+ (+40%y/y) in 2028. Given almost unanimous supplier agreement, we struggle to get above \$75bn of DRAM WFE in 2027.

The Numbers: We forecast \$120bn+ of DRAM WFE over 2026/27, as we model \$53bn (+70% y/y) in 2026 and \$70bn (+33% y/y) in 2027. We're curious how the industry gets to \$75bn+ in 2027, but given unanimous supplier agreement, we see risk skewed to the upside for 2027.

Suppliers are implying \$75bn+ in 2027. SPE companies almost unanimously agree that DRAM WFE growth will outpace total WFE in 2027. These comments imply DRAM WFE north of \$75bn (40% y/y growth).

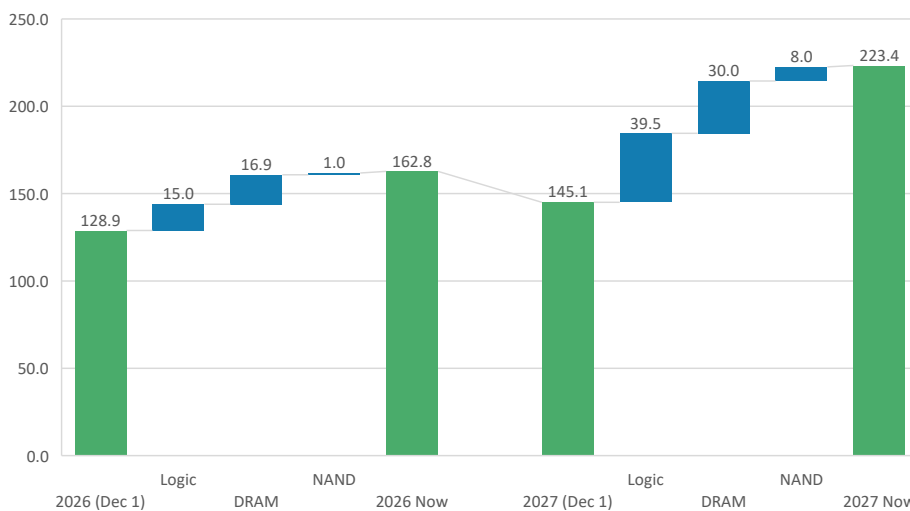
- "We expect the growth drivers next year to be largely the same as they are this year, led by growth in DRAM WFE, followed by leading-edge foundry and logic, followed by NAND." (LAM Research – July 29th, 2026)
- We expect DRAM to continue leading the market, and we also have high expectations for it." (Kokusai Electric - August 6th, 2026)
- "The fastest-growing parts of the market are the leading-edge foundry-logic, DRAM, and advanced packaging. We said that's around 80% of the growth in wafer fab equipment spending this year. And then we'll see a similar profile in 2027." (Applied Materials - August 13th, 2026)

2026 has been a big year. DRAM WFE has tracked far above our expectation as our new 2026 forecast of \$53bn is 47% higher than what we saw going into the year ([Semiconductor Capital Equipment: 4Q'25 WFE update, New DRAM WFE Paradigm \(1 Dec 2025\)](#)). The DRAM suppliers called out "cleanroom constraints" as a limitation to supply growth, but supply has been added via creative ways;

- **Micron – PSMC:** Micron completed its [acquisition of PSMC's Tongluo P5 site in March 15th](#) and we believe the company has been able to expedite tool installation (\$4-5bn incremental in 2026).
- **Brownfield:** We estimate brownfield DRAM WFE grew 48% in 2026 as Micron accelerated node transitions to 1Gamma.
- **Samsung P4:** We believe more cleanroom capacity at Samsung's P4 fab has been allocated to DRAM and see 85kwpw of wafer starts exiting 2026.

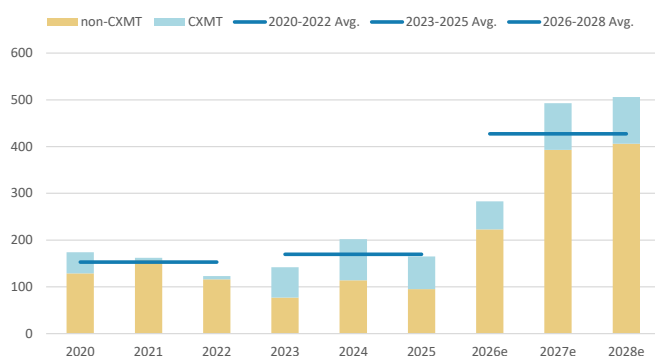
The DRAM makers have found creative ways to meet unprecedented demand with unprecedented WFE expedites.

Exhibit 27: Our 2026 DRAM WFE forecast has come up by approx. 50% since Dec 1st as manufacturers have found ways to expedite tool deliveries and free up cleanroom.



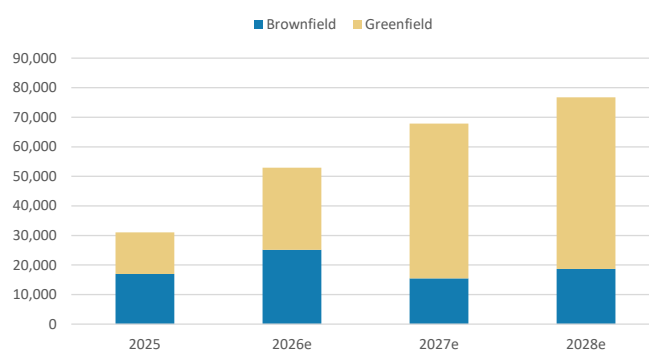
Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 28: We model greenfield adds to average 427kwpm during 2026-28



Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 29: Following 48% brownfield growth in 2026, we model a deceleration in 2027



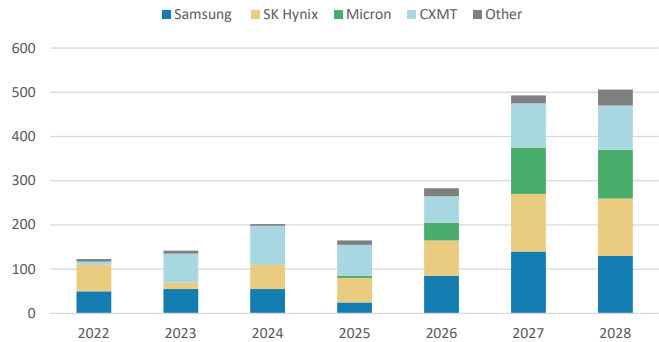
Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

\$120bn+ over 2 years. We're not making a strong call by modeling a deceleration, we're saying "there was a lot of pull-forward into 2026 which makes the 2027 comp significantly higher." We model \$120bn+ of DRAM WFE over 2026/2027 but see risk skewed to the upside. Where could we be wrong on 2027?

- **Greenfield:** We model 493kwpm of greenfield in 2026, almost double 2026 (283kwpm) and almost triple 2025 (165kwpm).
- **Brownfield:** Given node migrations were pulled forward into 2026, we model brownfield to decelerate significantly from 2026 (2027 \$15.5bn, 2026 \$25bn).

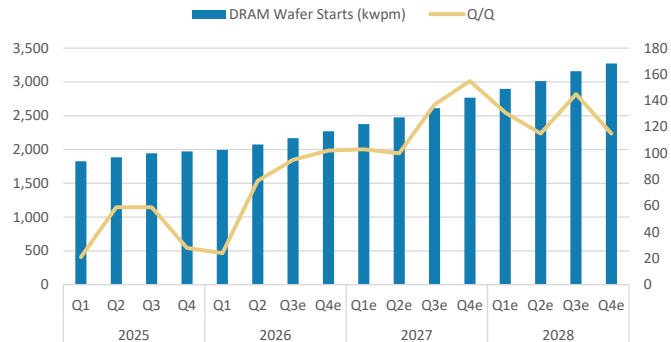
We would be surprised if greenfield tracks above our forecast. We could be missing some brownfield investment as we do model a deceleration.

Exhibit 30: 2027/28 greenfield to be driven by all 4 players and others (Nanya, Winbond, JHICC)



Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 31: We now model DRAM Wafer starts to surpass 3,000kwpm in Q2 2028 (Q3 2026 2,169kwpm)



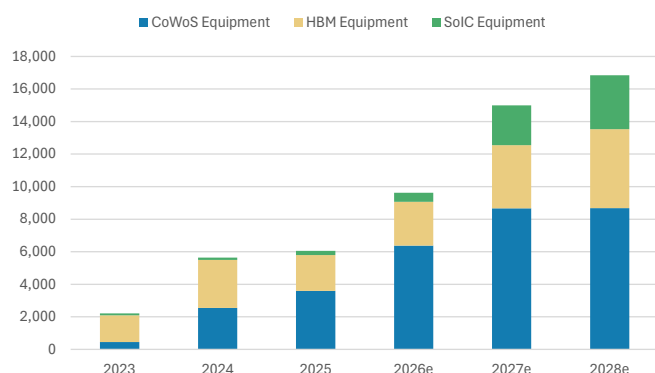
Source: TrendForce, Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Thought #3: Advanced Packaging

Our View: AMAT, LAM, and KLA have all guided for Advanced Packaging growth of 70%+ in 2026. We think this growth was driven by CoWoS adds, and we think 2027 should be another year where AP tracks ahead of WFE growth, led by SoIC and HBM.

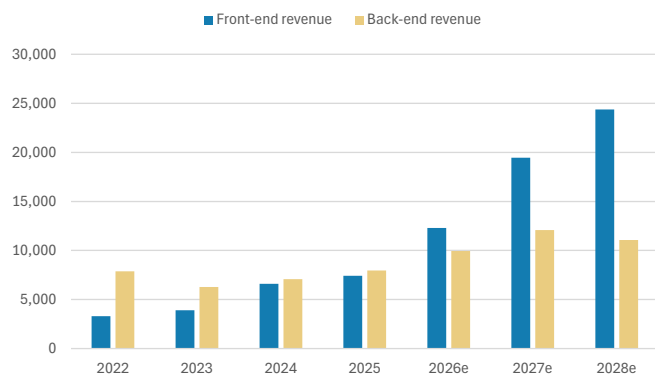
The Numbers: Our Advanced Packaging numbers are largely unchanged as we model CoWoS, HBM, and SoIC equipment demand to grow 59%/56% in 2026/27.

Exhibit 32: We model CoWoS, HBM, and SoIC equipment demand to grow 59%/56% in 2026/27



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 34: Front-end related AP revenue continues to outpace back-end



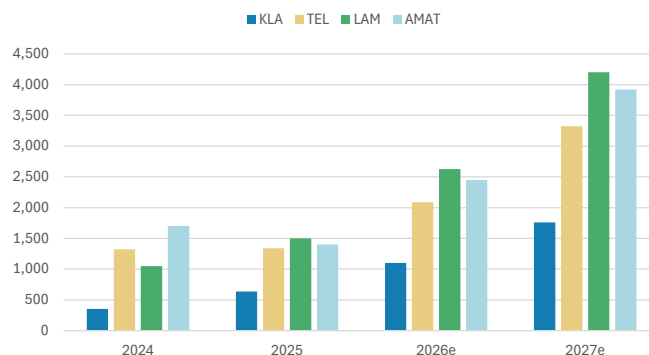
Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 33: 2026 was driven by CoWoS adds, we expect 2027 to be driven by HBM and SoIC.



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 35: We model AMAT, LAM, KLA, and TEL to all be growing advanced packaging revenue approx. 70%.



Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Shane Brett; Charlie Chan; Nicole Kozhukhov; Joseph Moore; Nigel van Putten; Amelia M Scicluna; Lee Simpson; Suzune Tamura, CFA; Ella Tulchinsky; Mason Wayne; Henry Zhao.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: Advanced Energy Industries Inc., Advantest, Applied Materials Inc., DISCO, Ebara, Intel Corporation, KLA Corp, KOKUSAI ELECTRIC, Lam Research Corp, Lasertec, MKS Inc., Nikon, ONTO Innovation Inc, SK hynix, Teradyne Inc, Tokyo Electron, TSMC, Ulvac, Ushio.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Advanced Energy Industries Inc., Camtek, Intel Corporation, KOKUSAI ELECTRIC, MKS Inc., Nova Ltd, ONTO Innovation Inc, Samsung Electronics, SK hynix, Tekscend Photomask.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Advanced Energy Industries Inc., Applied Materials Inc., Camtek, Intel Corporation, JEOL, MKS Inc., Nova Ltd, SK hynix, Tekscend Photomask, Tokyo Electron.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Energy Industries Inc., Advantest, Applied Materials Inc., Camtek, DISCO, Ebara, Intel Corporation, JEOL, KLA Corp, KOKUSAI ELECTRIC, Lam Research Corp, Lasertec, MKS Inc., Nikon, Nova Ltd, Samsung Electronics, SCREEN Holdings, SK hynix, Tekscend Photomask, Teradyne Inc, Tokyo Electron, TSMC, Ulvac, Ushio.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Applied Materials Inc., Intel Corporation, MKS Inc., Samsung Electronics, Tokyo Electron, TSMC.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Energy Industries Inc., Advantest, Applied Materials Inc., Camtek, Ebara, Intel Corporation, JEOL, KLA Corp, KOKUSAI ELECTRIC, Lam Research Corp, Lasertec, MKS Inc., Nikon, Nova Ltd, ONTO Innovation Inc, Samsung Electronics, SCREEN Holdings, SK hynix, Tekscend Photomask, Teradyne Inc, Tokyo Electron, TSMC, Ulvac, Ushio.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Advanced Energy Industries Inc., Applied Materials Inc., Intel Corporation, MKS Inc., Samsung Electronics, SK hynix, Tokyo Electron, TSMC.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

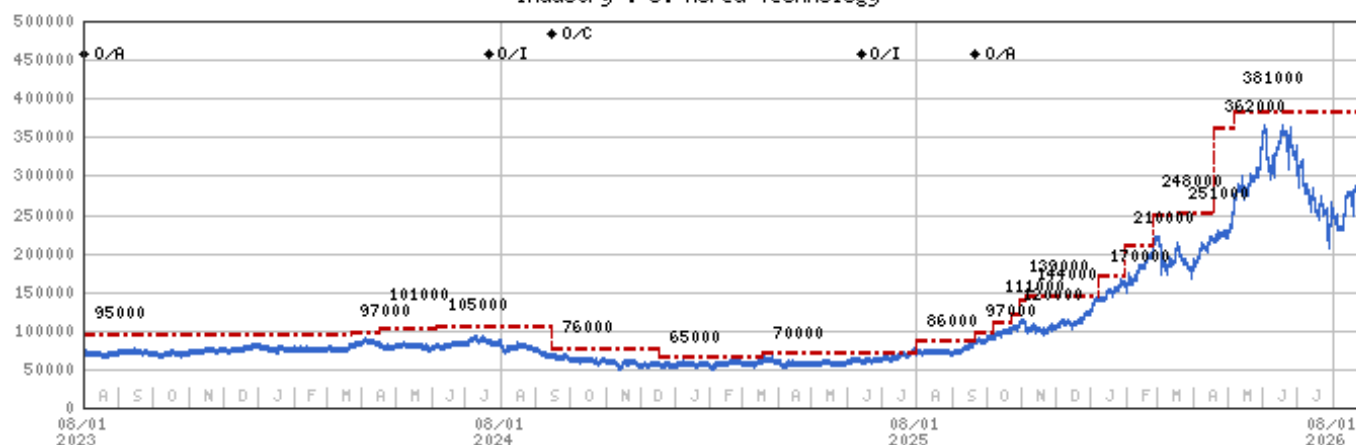
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Samsung Electronics (005930.KS) - As of 08/24/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

Price Target History: 6/6/21 : 98000; 8/12/21 : 89000; 9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000;
6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000; 9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000;
4/16/24 : 101000; 6/6/24 : 105000; 9/15/24 : 76000; 12/18/24 : 65000; 3/19/25 : 70000; 8/1/25 : 86000; 9/21/25 : 97000;
10/8/25 : 111000; 10/23/25 : 120000; 10/30/25 : 139000; 11/5/25 : 144000; 1/8/26 : 170000; 1/30/26 : 210000; 2/24/26 : 248000;
3/18/26 : 251000; 4/19/26 : 362000; 5/6/26 : 381000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

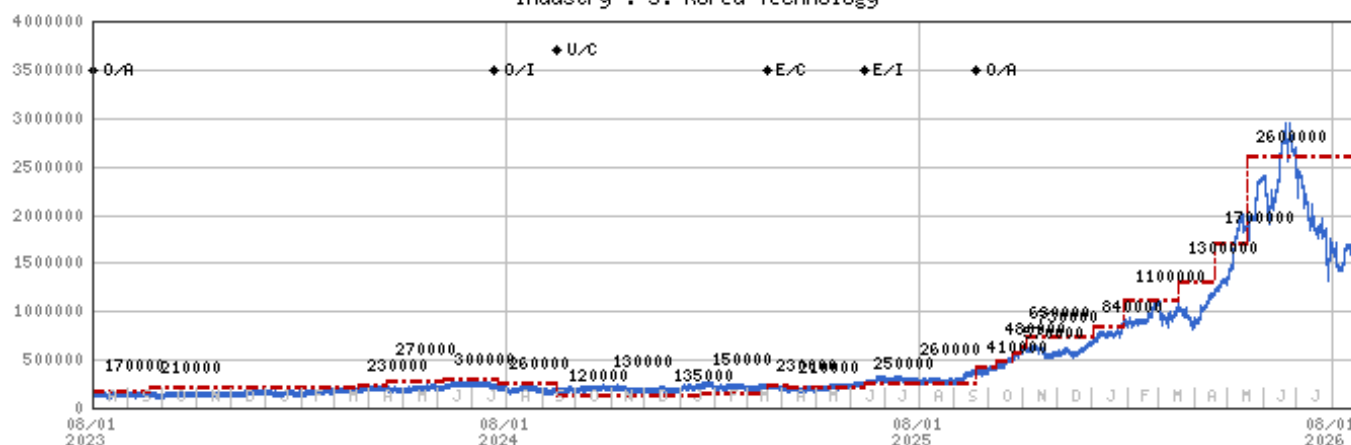
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

SK hynix (000660.KS) - As of 08/24/26 GMT in KRW
Industry : S. Korea Technology

Stock Rating History: 8/1/21 : O/I; 8/12/21 : U/C; 12/3/21 : E/C; 2/11/22 : O/C; 7/22/22 : E/C; 10/4/22 : O/A; 7/21/24 : O/I;
9/15/24 : U/C; 3/19/25 : E/C; 6/13/25 : E/I; 9/21/25 : O/A

Price Target History: 6/8/21 : 156000; 8/12/21 : 80000; 9/15/21 : 88000; 12/3/21 : 110000; 12/23/21 : 125000; 1/24/22 : 130000;
1/28/22 : 136000; 2/11/22 : 155000; 3/18/22 : 150000; 4/27/22 : 130000; 6/10/22 : 120000; 7/5/22 : 110000; 7/22/22 : 105000;
10/4/22 : 130000; 12/7/22 : 120000; 3/21/23 : 110000; 5/30/23 : 140000; 7/7/23 : 170000; 9/21/23 : 210000; 3/22/24 : 230000;
4/16/24 : 270000; 6/6/24 : 300000; 7/25/24 : 260000; 9/15/24 : 120000; 10/24/24 : 130000; 12/18/24 : 135000; 1/20/25 : 150000;
3/19/25 : 230000; 4/7/25 : 210000; 6/13/25 : 250000; 7/24/25 : 260000; 9/21/25 : 410000; 10/8/25 : 480000; 10/23/25 : 570000;
10/29/25 : 630000; 11/5/25 : 730000; 1/2/26 : 840000; 1/30/26 : 1100000; 3/18/26 : 1300000; 4/19/26 : 1700000; 5/18/26 : 2600000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Ratings/Industry View
Stock Ratings: Overweight (O) Equal-weight (U) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (<https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/>

[MSICPL_Morgan_Stanley_Research_Audit_Report.pdf](#)).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member

of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The following companies do business in countries which are generally subject to comprehensive sanctions programs administered or enforced by the U.S. Department of the Treasury's Office of Foreign Assets Control ("OFAC") and by other countries and multi-national bodies: Samsung Electronics.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Semiconductor Capital Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/21/2026)
Shane Brett		
Advanced Energy Industries Inc. (AEIS.O)	O (07/22/2026)	\$285.83
Applied Materials Inc. (AMAT.O)	E (05/18/2026)	\$492.32
Camtek (CAMT.O)	E (12/01/2025)	\$146.81
KLA Corp (KLAC.O)	O (01/15/2026)	\$183.99
Lam Research Corp (LRCX.O)	O (05/18/2026)	\$314.00
MKS Inc. (MKSI.O)	O (08/04/2024)	\$279.11
Nova Ltd (NVMI.O)	E (12/01/2025)	\$376.99
ONTO Innovation Inc (ONTO.N)	O (06/14/2026)	\$293.33
Teradyne Inc (TER.O)	E (07/30/2025)	\$375.74

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Semiconductor Production Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/24/2026)
Suzune Tamura, CFA		
Advantest (6857.T)	O (01/17/2024)	¥34,500
DISCO (6146.T)	O (07/07/2023)	¥61,420
Ebara (6361.T)	O (12/04/2024)	¥4,960
JEOL (6951.T)	E (07/08/2025)	¥7,134
KOKUSAI ELECTRIC (6525.T)	O (12/19/2025)	¥8,280
Lasertec (6920.T)	U (07/28/2025)	¥34,810
Nikon (7731.T)	U (08/08/2023)	¥1,981
SCREEN Holdings (7735.T)	O (11/13/2025)	¥13,150
Tekscend Photomask (429A.T)	E (11/18/2025)	¥3,710

Tokyo Electron (8035.T)	O (12/19/2025)	¥55,150
Ulvac (6728.T)	E (01/17/2024)	¥7,572
Ushio (6925.T)	O (01/05/2026)	¥3,872

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley